

TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature, on *which* lead, a clinician can trust. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S , and a target lead ℓ , two closed-form numbers derived from one truncated SVD of the segment’s empirical rank-3 spatial (“dipolar”) sub-matrix—an *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target’s low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$ (its noise/observed-residual gain). The per-lead criterion is the classical estimability condition specialized to ECG, but made *graded*: on PTB-XL the map ranks target leads on a continuum—e.g. from limb leads, anterior precordial ST (V1–V4) is not identifiable while lateral V5/V6 are near-identifiable—a call a physiological precordial/limb split cannot make. It is stable to the truncation tolerance for well-posed configurations and rank-sensitive (with reported sensitivity) for ill-posed ones. We attach distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression, strict fold discipline, honestly stated within-group coverage), and give a reconstructor-*invariant* safety certificate: the total ST-threshold error floor is near-invariant across reconstructors while the false-positive/false-negative split is a design choice. The map turns “trust the reconstruction” into a per-feature, per-lead, calibrated statement.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead’s dipolar morphology is even identifiable, and how well-conditioned is its*

recovery? Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis M_s (top-3 singular vectors of the population lead covariance). We stress that M_s is a *population-estimated (PCA)* object: on real PTB-XL it shares two directions with the classical vectorcardiographic (inverse-Dower) dipole but its third direction differs materially, so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 4).

Related work and contributions. Classical work reconstructs missing leads (regression / VCG synthesis [3, 4, 5], deep models [1, 2]) or *selects* optimal leads [6]; whether a given target is even recoverable is the classical estimability question [7]. None attaches a graded, per-target-lead recoverability certificate with calibrated per-record error to a reconstruction. We do not claim the underlying identity as new. Our contributions are: (1) a *graded*, per-target-lead recoverability map—continuous $\eta_{s,\ell}(S)/\kappa_{s,\ell}(S)$ from a single truncated SVD that *rank* target leads (not a binary rank test), over an *empirically-estimated* M_s that differs 43–55° from the textbook inverse-Dower dipole, with a truncation-tolerance sensitivity analysis and record-bootstrap CIs (Sec. 2); (2) a calibration layer attaching distribution-free intervals (a sound application of conformalized quantile regression + Mondrian [8], not a new method) to the map per feature and lead, with strict fold discipline and honest within-group (not per-diagnosis) coverage (Sec. 3); (3) a reconstructor-invariant safety certificate on PTB-XL—the total ST-threshold error floor is near-invariant across reconstructors while the false-positive/false-negative split is a design choice—showing anterior precordial ST is uncertifiable from limb leads (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Model. Within segment s write the population 12-lead vector as $L_s = \mu_s + M_s d + r_s$, where $M_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, d the dipole coordinates, and r_s the non-dipolar residual. Observing S gives $y_S = \text{Sel}_S L_s + n$. Let $M_{s,S} = \text{Sel}_S M_s$ have truncated SVD at relative tolerance ϱ ; $M_{s,S}^+$ its truncated pseudo-inverse and $P_{\text{obs}} = M_{s,S}^+ M_{s,S}$ the projector onto the observed dipole coordinate directions.

Proposition 1 (Per-target-lead dipolar identifiability and conditioning). *The mean-centred dipolar estimate $\hat{\mathbf{L}}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{y}_S - \boldsymbol{\mu}_{s,S})$ satisfies, for each target lead ℓ with $\mathbf{e}_\ell$ the ℓ -th unit vector:*

- (i) *Identifiability. Define $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$. If $\eta_{s,\ell}(S) = 0$, the dipolar component of lead ℓ is identifiable from S : $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ is determined by $\text{Sel}_S \mathbf{M}_s \mathbf{d}$. If $\eta_{s,\ell}(S) > 0$, there is a dipole direction unobserved by S that changes lead ℓ ; its dipolar component is not identifiable at any SNR.*
- (ii) *Conditioning. On the identifiable part, the error from observation noise and from the observed non-dipolar residual is amplified into lead ℓ by $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$: $|\mathbf{e}_\ell^\top (\hat{\mathbf{L}}_s - \boldsymbol{\mu}_s - \mathbf{M}_s \mathbf{P}_{\text{obs}} \mathbf{d})| \leq \kappa_{s,\ell}(S) (\|\text{Sel}_S \mathbf{r}_s\| + \|\mathbf{n}\|)$.*

The global $\kappa_s(S) = \|\mathbf{M}_s \mathbf{M}_{s,S}^+\|_2 = \max_\ell \kappa_{s,\ell}(S)$ is a configuration-level worst case. Both $\eta_{s,\ell}$ and $\kappa_{s,\ell}$ depend on the estimated $\mathbf{M}_s$ and on ϱ ; near-rank-deficient S are reported with an ϱ -sensitivity sweep and record-bootstrap confidence intervals.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [7]) specialized to the per-lead linear functional $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$: $\eta_{s,\ell}(S) = 0$ iff that functional lies in the row space of $\mathbf{M}_{s,S}$, and $\kappa_{s,\ell}$ is its coefficient norm. We claim no novelty for this identity. The contribution (Sec. 4) is its use as a *graded*, per-target-lead, *empirically-estimated* recoverability map: a continuous η that ranks target leads (not a binary rank test), a subspace $\mathbf{M}_s$ that differs materially from the textbook inverse-Dower dipole, and per-record calibrated error rates (Sec. 3) attached to it—quantities not read off lead geometry.

The non-dipolar residual (no independence claim on real data). Let $\mathbf{m}_s(\mathbf{y}_S) = \mathbb{E}[\mathbf{r}_s \mid \mathbf{y}_S]$ and $\mathbf{u}_s = \mathbf{r}_s - \mathbf{m}_s(\mathbf{y}_S)$, so $\mathbb{E}[\mathbf{u}_s \mid \mathbf{y}_S] = \mathbf{0}$. We do *not* assume $\mathbf{u}_s \perp \mathbf{y}_S$. The Bayes-optimal reconstruction error of the residual is

$$\inf_{\hat{\mathbf{r}}} \mathbb{E} \|\mathbf{r}_s - \hat{\mathbf{r}}(\mathbf{y}_S)\|^2 = \mathbb{E} [\text{Var}(\mathbf{r}_s \mid \mathbf{y}_S)].$$

Part of $\mathbf{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\mathbf{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—not declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only). *If the data are generated so that a component $\mathbf{u}$ is statistically independent of $\mathbf{y}_S$ (an explicit synthetic construction), then $p(\mathbf{u} \mid \mathbf{y}_S) = p(\mathbf{u})$, the Bayes estimate of $\mathbf{u}$ is its prior mean, and $\mathbb{E} \|\hat{\mathbf{u}} - \mathbf{u}\|^2 \geq \text{Var}(\mathbf{u})$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

3. DISTRIBUTION-FREE CALIBRATION

Predictable-residual intervals. For each pre-registered group $g = (S, s, \ell)$ we train a quantile regressor of the target lead’s off-dipole residual on the observed leads (gradient-boosted pinball loss) and apply conformalized quantile regression [8] with a Mondrian (per-group) correction. Fold discipline is strict and leakage-free: basis and quantile model on folds 1–7, conformal calibration on fold 9, a single evaluation on fold 10. We report *within-group marginal coverage* under exchangeability (not per-example conditional coverage), interval width, and group sizes.

Off-dipole flag. We threshold the off-dipole energy at the one-sided $(1 - \alpha)$ conformal quantile over faithful reconstructions; the false-flag rate is $\leq \alpha$ for a fresh reconstruction *exchangeable with the calibration distribution*—not distribution-free under an arbitrary shift.

4. EXPERIMENTS

Data. PTB-XL [9] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [10].

Reproducibility. Code, fixed seeds, an exact environment lock, and a one-command pipeline are released¹; every number in the tables is regenerated from result JSON (no hand-typed values). The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, Adam 2×10^{-4} , MSE on masked leads under random lead masks); the residual predictor is a gradient-boosted quantile regressor (pinball loss). The map is evaluated on records disjoint from the folds used to estimate $\boldsymbol{\mu}_s, \mathbf{M}_s$.

4.1. The recoverability map on PTB-XL

We estimate $\mathbf{M}_s$ (rank 3) per segment on the training folds and evaluate the map on 1,200 NORM records disjoint from the estimation folds (per-lead η, κ with 200-sample bootstrap CIs, truncation tolerance $\varrho=10^{-2}$; Fig. 1). The map is *graded and per target lead*: a single observed lead (e.g. Lead I) leaves every other lead’s dipolar component unidentifiable ($\eta > 0$; rank 1); a dipole-spanning triplet {I,II,V2} or the five-lead {I,II,V1,V3,V5} makes all targets identifiable ($\eta \approx 0$ to machine precision, since S then spans the rank-3 subspace). The value over a binary rank test is the *gradient* it resolves: for limb-6 the ST identifiability falls smoothly from strongly unidentifiable anterior leads (V2 $\eta=0.712$, V3 0.553, V1 0.330, V4 0.268) to *near-identifiable* lateral leads (V5 0.084, V6 0.027). So the honest a priori warning is specific: **anterior precordial ST (V1–V4) is not identifiable from limb leads**—the direction of the estimated subspace it needs is unobserved—while lateral V5/V6 are largely recov-

¹Anonymized for review; released on acceptance.

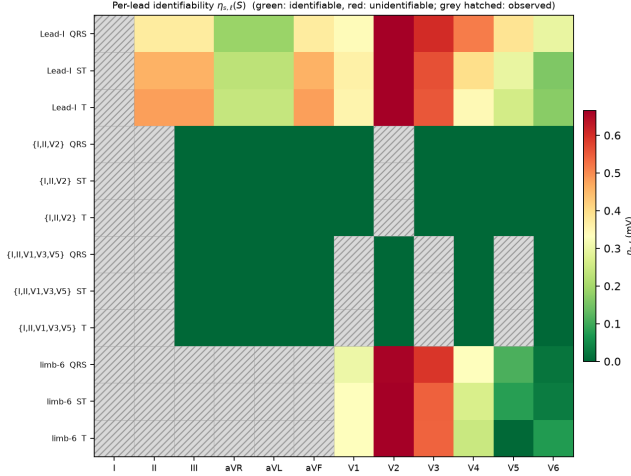


Fig. 1. Per-lead identifiability $\eta_{s,\ell}(S)$ across configurations and segments (green: identifiable, red: unidentifiable; grey hatched: observed). For limb-6 the anterior precordial leads are unidentifiable while lateral V5/V6 grade toward identifiable.

Table 1. Limb-6 $\rightarrow$ anterior-precordial ST reconstruction, 1498 fold-10 records. FP = false-positive, FN = false-negative ST-threshold crossings (%). The *total* (FP+FN) is near-invariant—the certified floor; the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	ST err (mV)
dipolar	32.5	17.6	50.1	0.076
ridge	5.3	42.3	47.6	0.059
OLS	1.1	46.0	47.1	0.081

erable. A physiological binary “precordial vs. limb” split cannot make this call; the empirical graded map does.

Certificate-level safety is reconstructor-invariant at the floor. The warning is actionable but must be stated on a reconstructor-invariant quantity. On 1498 delineated fold-10 records we reconstruct anterior precordial ST from limb-6 with three reconstructors and count ST-threshold events ($|ST| > 0.1$ mV crossings)—false positives (a crossing in the reconstruction, not the truth) and false negatives (vice versa; Table 1). The *total* wrong-event rate is near-invariant across reconstructors (47.1–50.1%, at mean ST error 0.059–0.081 mV): this floor is the cost of the certified unidentifiability. What the reconstructor *does* choose is the false-positive/false-negative split—dipolar leans to false positives (32.5%/17.6%), OLS to false negatives (1.1%/46.0%)—a design trade-off, not a certified number. We therefore report the whole matrix and anchor the certificate on the invariant floor, not a single cell.

Truncation-tolerance sensitivity. Rank/ κ of well-posed

configurations are stable across $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$ (e.g. {I,II,V1,V3,V5} is rank 3 throughout), while near-rank-deficient sets are sensitive: on synthetic data {V1,V2,V3} is rank 2 at $\varrho=10^{-2}$ but rank 3 with $\kappa \sim 200$ at $\varrho=10^{-4}$. We therefore report a single κ only for well-conditioned configurations and give the ϱ -sweep and bootstrap CIs otherwise; a single 10^4 or 10^5 figure for a degenerate set is not meaningful.

4.2. Calibrated intervals for the predictable residual

For the empirically predictable residual we train a gradient-boosted quantile regressor of each target lead’s off-dipole residual on the observed leads and apply Mondrian conformalized quantile regression [8] per (S, s, ℓ) group, with strict fold discipline (basis and model on folds 1–7, calibration on fold 9, a single evaluation on fold 10; nominal $1-\alpha=0.90$). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.88–0.95; e.g. {I,II,V1,V3,V5} QRS/V2 0.92, record-bootstrap CI [0.89, 0.95], mean width 0.165 mV).

What is and is not guaranteed. Mondrian CQR gives *within- (S, s, ℓ) -group marginal coverage* under exchangeability—not coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.80 ($n=40$), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

4.3. Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: per-timepoint waveform RMSE on the same 800 held-out fold-10 records, same segment indices, record-bootstrap CIs. We include a *strong neural baseline*: an arbitrary-mask conditional 1-D U-Net trained with random lead masks (so one model serves any configuration, no target-lead leakage), the masked-reconstruction architecture the reduced-lead literature uses [1, 5].

On a dipole-spanning set the methods form a monotone ladder—prior-mean $\rightarrow$ dipolar (Tier I) $\rightarrow$ per-segment ridge (Tier I+II) $\rightarrow$ U-Net—confirming there is recoverable non-dipolar content (the calibrated Tier II layer), yet the strong neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.168 mV on {I,II,V1,V3,V5}). On **limb-6** the U-Net still *lowers* error substantially over the prior mean

Table 2. Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol (800 test records). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = strong arbitrary-mask neural baseline. Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error, the gap being the unobserved dipolar coordinate $\eta > 0$ flags. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.471	0.234	0.220	0.181
{I,II,V1,V3,V5}	0.484	0.186	0.168	0.164
limb-6	0.484	0.362	0.346	0.239

(QRS 0.484 $\rightarrow$ 0.239)—it recovers the predictable, largely population-correlated structure—but plateaus at $1.46\times$ its spanning-set error and no reconstructor closes that gap. The residual gap is exactly the *unobserved dipolar coordinate* the map’s $\eta > 0$ marks as unidentifiable: capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing that coordinate. We therefore scope the certified claim to the dipolar component, not to a blanket “not recoverable.”

Empirical, not physical, subspace. On real PTB-XL the estimated M_s shares two directions with the classical inverse-Dower vectorcardiographic space [3, 4] (principal angles 2–6°) but its third direction differs materially (max angle 43–55°, bootstrap CIs excluding small angles). We therefore treat M_s as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all certificate quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment.

5. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a graded per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It tells a clinician which lead configuration can support a decision, e.g. that anterior precordial ST cannot be read from limb leads while lateral V5/V6 can, and reports the reconstructor-invariant error floor that any recovery must pay.

6. REFERENCES

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